SandEE: AN EDUCATIONAL 2D OPEN WORLD SURVIVAL CRAFTING GAME FOR LUA AND PYTHON

A Thesis Proposal
Presented to the Faculty of the
Information and Communications Technology Program
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In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Science

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **SandEE: An Educational 2D Open World Survival Crafting Game for Lua and Python**

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INTRODUCTION

In recent times, especially in information communication technology, programming has been the focus of computing education. The recent advent of several tools has made coding more accessible than ever (Becker, 2023). Programming has been a challenge for learners. The proliferation of digital technologies has made programming skills increasingly important across various disciplines. This shift extends beyond traditional computer science domains (Lin, et.al., 2025). According to industry reports, the demand for Lua developers has been steadily increasing in recent years. As more companies turn to Lua for its speed, simplicity, and versatility, the need for skilled Lua developers has never been higher. In particular, the gaming industry has seen a rise in Lua-related job openings, with many companies looking for developers proficient in Lua scripting (Crudu, 2024). The Lua scripting language is a fast, procedural programming language. The language was designed to be highly customizable with simple API in the C programming language so that programmers could easily make changes to fit their needs (Ierusalimschy, et al., 2025). While Python is a popular programming language created by Guido Van Rossumed released in 1991. It has many applications including web development, software development, mathematics, and system scripting. Python has a simple syntax that allows developers to write programs with fewer lines compared to other programming languages (Python, 2021). Additionally, Python as a scripting language has been steadily gaining traction since its inception. Its simplified syntax allows learners to easily learn and grasp the coding structure of the scripting language (Python, 2021).

In a survey conducted to ICT and STEM Senior High School students of Yverdon De Pestallozi School Inc., a total of 104 respondents participated, consisting of 28 students from Grade 11 ICT (26.9%), 24 from Grade 11 STEM (23.1%), 30 from Grade 12 ICT (28.8%), and 24 from Grade 12 STEM (23.1%). The results revealed that most students demonstrated low familiarity with the Lua programming language, suggesting that it is relatively new to many respondents. Despite this, the majority expressed positive attitudes toward learning Lua through game-based activities, with 78 (76%) agreeing that such an approach would improve their understanding of programming concepts and 83 (79.8%) showing motivation to explore Lua within game environments. On the other hand, Python received generally favorable feedback, with 72 (70%) of participants finding its syntax and structure easy to understand and applicable to programming tasks. Overall, the findings indicate that while students are less experienced with Lua compared to Python, they show strong enthusiasm and willingness to learn it through interactive and game-oriented methods that enhance engagement and learning outcomes. As a result, introducing a gamified approach of learning Lua and Python is applicable to the students at the school.

SandEE is an educational 2D open world survival crafting game for Lua and Python. It introduces several programming concepts related to Lua and Python. Furthermore, SandEE also includes gameplay that can serve as an exercise to advanced users of Lua and Python. The proposed study offers a supplementary learning approach for Lua and Python by slowly introducing its syntax and structure. The game embeds interpreters for Lua and Python in which code that is run impacts the game environment to show visual feedback, as well as to provide a realistic basis to the player as the interpreters will give information accurate to its real use-case. Progression is achieved through the mission system, which guides the player to progress in learning. These missions will eventually lead to rewards to further gratify the player and give them feedback on their progress.

Background of the problem

The rapid advancement of technology has significantly transformed educational methodologies. Traditional learning, which often emphasizes theoretical concepts and static exercises, struggles to fully engage students or prepare them for real-world problem-solving. In contrast, educational games have emerged as an innovative approach to learning, combining interactivity and engagement with skill development. Popular examples like Duolingo demonstrate how gamified systems can make learning both effective and enjoyable. Game-based learning in programming has shown the potential to enhance logical reasoning, problem-solving, and creativity through active participation.

To better understand the learning preferences of ICT and STEMS students, a survey was conducted among the grade 11 and 12 ICT and STEM students of Yverdon De Pestallozi School Inc. The results revealed that most respondents had limited familiarity with the Lua programming language, with the majority rating their understanding between 1 and 2 on a 4-point scale. However, despite this lack of experience, 83 (70%) of respondents rated 3 or 4 when asked how helpful and motivating it would be to learn Lua through game programming. Additionally, a large portion of students were familiar with survival games, which suggests that integrating coding challenges within a game environment would appeal to their interests. For comparison, students showed higher familiarity with Python, averaging between 3 and 4 in both ease of use and coding level. Yet even those comfortable with Python expressed strong interest in learning programming concepts through interactive, game-based methods. These findings indicate that students are more receptive and motivated when learning involves playing, which shows what happens inside the game, experimentation, and engagement rather than passive listening and reading books.

To address these findings, the proposed game introduces an immersive, game-based platform that integrates survival and crafting mechanics within an open-world sandbox-type environment. Through hands-on coding challenges embedded in gameplay, it aims to foster critical thinking, logical reasoning, and practical programming skills. This approach seeks to make programming educational, creative, experiential, contextual, and enjoyable, bridging the gap between theoretical instruction and real-world application.

General Problem

How to develop a game to supplement the high interest in learning Lua and Python through a game?

Despite low familiarity with ICT and STEM students, survey results have shown that interest in learning Lua and Python is high. Furthermore, the results have also shown an increasing interest in learning these languages using a more interactive and engaging approach through a game.

Specific Problem

- **How to develop a game that enhances programming skills by providing hands-on learning where concepts are learned through experience?**

  Many ICT and STEM students have shown a strong interest in learning programming languages such as Lua and Python, particularly through interactive and engaging methods. Based on the survey conducted, there is a clear inclination toward learning through game-based experiences that allow the practical use of programming concepts. Therefore, this study focuses on developing a game that enhances programming skills by providing hands-on learning where concepts are learned through experience, serving as an innovative approach to support growing interest in programming.

- **How to develop a game that provides direct visual feedback as players learn and apply programming concepts?**

  Students often show greater engagement and understanding when they can immediately see the results of their actions while learning. In programming education, providing direct visual feedback allows learners to observe how their inputs affect outcomes in real time, creating a more interactive and immersive experience. Therefore, this study aims to develop a game that provides direct visual feedback as players learn and apply programming concepts, promoting better comprehension and active participation through visual interaction.

- **How to develop a game that realistically responds to player errors by embedding Lua and Python interpreters within the system?**

  To create a more authentic and educational programming experience, it is important for a game to respond realistically to player inputs, including coding errors. Embedding Lua and Python interpreters within the system enables real-time evaluation of player code, allowing the game to simulate how actual programming environments handle mistakes.

- **How to develop a game that introduces the programming structure and syntax of Lua and Python?**

  Learning programming languages such as Lua and Python requires a clear understanding of their syntax and structural rules. Introducing these foundational concepts through an interactive approach can make learning more engaging and effective.

Overview of the current state of the technology

In recent years, advancements in teaching coding through educational and engaging means have been accomplished. There are ways, including game-based learning and digital courses, that provide plenty of valuable knowledge on how to code to solve problems and perform tasks.

Philippine Coding Camp (2023)

Such examples include the **Philippine Coding Camp**. These are digital learning bootcamps that primarily focus on teaching students of any age by making the courses simpler, welcoming, and open to beginners and experienced learners alike. Both provide structured courses that teach the fundamentals and concepts of programming, which they can later apply to what they have learned through exercises and challenges. They use platforms such as Roblox as a supplement and make learning more engaging. As learners progress through the courses, new concepts are slowly and simply introduced for them to better understand and retain what they are learning.

While these courses are structured, there are limitations that can be addressed. Such as the focus is only centered on the topics of the course. By making the process more hands-on, it removes the restriction of complying with the structure of the course and letting the learners learn in their own way with similar guidance.

Logiscool (2025)

Logiscool is a similar coding camp established worldwide and has a digital platform where they also teach digital courses. They provide programs where in each, they teach digital literacy, AI, and coding. They integrate gaming platforms such as Roblox and Minecraft into their learning curriculums to make learning more interesting. In addition, they also use artificial intelligence to support teaching. Part of the programs they teach are emerging languages such as Python and Lua. Python is taught within topics such machine learning, and Lua is primarily used in Roblox to create games and experiences. Logiscool's main target is younger users but are still open to everyone.

Logiscool's programs thoroughly teach its learners about the topics, provide and hones skills such as critical thinking and creativity. But by creating a more engaging way to teach, similarly to how they use games, learners can better learn and apply their skills into practical applications.

Hello World (2024)

Hello World is a game that primarily teaches Java and C#. It teaches fundamental concepts of these two-programming languages and integrates them into gameplay. Throughout Hello World, introductory lessons are provided, then the player must use what they have learned to progress. The game then adds the challenge of using different concepts to solve a given problem. It teaches through direct engagement with the languages since it is incorporated into the gameplay. The strength of the system comes from the interactive process of learning languages.

This study provides aims to recreate the process of learning through interaction with languages, but uses two emerging languages, Lua and Python. Java and C# are already commonly taught in the field, but there is a growing need in the industry that seeks developers with knowledge of Lua and Python.

Code Combat (2025)

CodeCombat is a digital gaming platform that primarily focuses on teaching fundamentals of programming and applying it into game scenarios. Ozaria is built from CodeCombat, where Python is one of the main focuses. They have a custom built-in engine and interpreter that helps beginners in learning how to code. This platform can be integrated in school settings, allowing students to share their projects with others and teachers to grade their work. CodeCombat works by teaching students first, then placing them in game scenarios where their knowledge is put to the test. Ozaria expands this by creating a storyline that progresses with the learner's skill and knowledge on the subject.

There are limitations to this, however. While this type of learning is more engaging, it lacks the structured format that courses provide. They teach first then send the learners into performing tasks. The study aims to bridge the gap between digital courses and gamified learning by blending the two ways of learning into a consolidated process. By merging the structured format of courses and the hands-on engagement that games provide, this creates a foundation for students to learn programming concepts in a unified form.

CodinGame (2025)

CodinGame is an online gaming platform that uses coding as its main method of playing. It challenges developers to use their skills in competing against others across different types of games. It encourages the application of what the user knows about programming. It features several programming languages such as Lua and Python, and developers must utilize their skills of the language throughout CodinGame's challenges and competitions. This type of learning helps developers see what they can still learn and improve to solve future challenges and tasks they may face.

CodinGame is limited since it primarily focuses on competition against other players. It doesn't provide an exact way to teach; it only points out what is lacking when it comes to knowledge. This type of learning is limited, hence creating a game that teaches, and rewards players gives a sense of accomplishment and progress that provides a better experience compared to the competitive nature of CodinGame.

Objectives of the study

General Objectives

To develop a game to supplement the high interest in learning Lua and Python through a game.

The proponents of the projects aim to develop a game with the goal of supplementing the high interest of learning the Lua and Python programming language through a game approach by integrating engaging gameplay with learning activities.

Specific Objectives

- **To develop a game that enhances programming skills by providing hands-on learning where concepts are learned through experience.**

  The proponents of the project aim to create an interactive learning environment where students can apply programming concepts directly within gameplay. Specifically, the game will encourage learners to solve coding challenges that simulate real-world scenarios, promote active problem-solving, and experiential learning rather than passive observation.

- **To develop a game that provides direct visual feedback as players learn and apply programming concepts.**

  The proponents of the project plan to use the game's environment to display visual feedback based on the user's coding input. Specifically, the player's actions and written code will directly affect the environment, allowing them to see how Lua and Python interpret their logic in real time. This feedback helps learners understand programming flow and identify errors through immediate, intuitive results.

- **To develop a game that realistically responds to player errors by embedding Lua and Python interpreters within the system.**

  The proponents of the project aim to ensure that the game reacts authentically to user errors by integrating Lua and Python interpreters. Specifically, this enables the application to detect and display syntax or runtime errors accurately, providing a realistic debugging experience. This approach helps students learn from their mistakes and develop stronger problem-solving and error-handling skills.

- **To develop a game that introduces the programming structure and syntax of Lua and Python.**

  The proponents of the project intend to integrate Lua and Python learning modules within the game. Specifically, the application will include coding activities that demonstrate syntax rules, control structures, and logical operations in both languages. This allows users to understand and practice programming concepts naturally while progressing through the game.

Scope and limitations of the study

- Main Menu
    - Start Game
        - This will redirect the user to the saves menu to select their desired save or start a new save.
    - Settings
        - This will redirect the user to the settings panel to adjust the game's sound volume.
    - Quit
        - This will close the game.
- Settings Panel
    - Volume Sliders
        - BGM Volume
            - Adjusts the volume for the background music of the game.
        - Effects Volume
            - Adjust the volume for the sound effects of the game.

- Saves Menu
    - Save Slots
        - This allows the user to manage their save states.
- Save Slot
    - Continue
        - Loads the chosen save file to continue its progress.
    - Delete
        - Deletes the chosen save file.

- Game Mechanics
    - Missions
        - A list of objectives will be shown to help track the user's progress.
    - Crafting
        - The user will be crafting various items that improve their current state or progress the missions provided.
    - Challenge Areas
        - Unlockable areas that serve to give major progress for the user, challenging their skills in Lua or Python programming to unlock more mechanics.
    - Automatica
        - Robot-like craftables that allow the user to automate certain processes by programming it through Lua, unlockable through the Lua-focused challenge areas.
    - Algorithmica
        - Allows the user to imbue regular items with automation by programming it through Python, unlockable through the Python-focused challenge areas.
    - LUA (Live Unit Application)
        - An advanced Automatica non-player character that introduces the user in the first few areas which helps the user during the Lua-focused challenge areas.

- Py-san
    - An advanced Algorithmica non-player character that introduces the user in the first few areas which helps the user during the Python-focused challenge areas.
- Autosave
    - The game will automatically save the player's progress during important parts of the game, as well as when the player quits.

- Desktop Application
    - HUD (Heads-Up Display)
        - Pause
            - Allows the user to temporarily pause the game, showing the Pause Menu Panel.
        - Pause Menu Panel
            - Resume
                - Resumes the current game.
            - Settings
                - Shows the settings panel.
            - Quit
                - Saves the game and directs the user to the Start Page.
        - Missions Panel
            - Shows the user the current missions they can do to progress.
        - Health Bar
            - Shows the user their current health.
        - Hotbar
            - Shows the user their current held items.
        - Inventory
            - Shows the user their extra space and a smaller crafting area.

- Code Editor Panel
    - A panel that prompts the user to code in Lua or Python to either progress in a challenge area or customize their Automatica and Algorithmica.
- Crafting Panel
    - A panel for more advanced crafting, unlockable when the user crafts the crafting bench.

- Mobile Application
    - HUD (Heads-Up Display)
        - Pause
            - Allows the user to temporarily pause the game, showing the Pause Menu Panel.
        - Pause Menu Panel
            - Resume
                - Resumes the current game.
            - Settings
                - Shows the settings panel.
            - Quit
                - Saves the game and directs the user to the Start Page.
        - Missions Panel
            - Shows the user the current missions they can do to progress.
        - Health Bar
            - Shows the user their current health.
        - Hotbar
            - Shows the user their current held items.
        - Inventory
            - Shows the user their extra space and a smaller crafting area.

- Code Editor Panel
    - A panel that prompts the user to code in Lua or Python to either progress in a challenge area or customize their Automatica and Algorithmica.
- Crafting Panel
    - A panel for more advanced crafting, unlockable when the user crafts the crafting bench.
- On-screen Joystick
    - Allows movement for mobile users.
- Interact Button
    - A general button that changes depending on the items near the player's character.

Limitations

- The game only runs on Windows 10 – 11 and Android 8.0 and above.
- The game only supports Lua and Python for its languages.
- The game only supports single player.

Review of related literature/studies/systems

Foreign Literature

According to Crudu (2024), Lua developers are in high demand because it is currently widely used in the programming industry. Lua is efficient, flexible, scalable, and cost effective to apply in any programming field. According to the TIOBE Index, Lua is currently ranked as the 32nd most popular programming language. Furthermore, in a survey conducted by Stack Overflow in 2024 it is among the top 10 most loved programming languages.

According to Becker (2022), Generating high-quality learning resources is very time consuming and typically requires a high level of expertise therefore, games have been used as a learning tool for centuries; they have become more prominent in educational settings. When it comes to learning, being focused and immersed can massively improve a student's experience.

According to Viduka (2021), Python is preferred by students and professionals alike due to its ease of use, availability, efficiency, portability, and extensibility. It is a common choice suitable for education. Many schools starting from primary to secondary, all the way to universities prefers teaching Python as primary programming languages.

Local Literature

Computer science software development evolves rapidly; innovative learning platforms transform how instructors teach and how aspiring developers learn programming. Several other solutions integrate interactive web based on learning programming languages. However, these methods of studying are not applicable to everyone. This is where a game-based solution can complement the traditional way of learning (Horlador, et., 2024)

The rapid development of computer science education signifies that traditional methods of learning are being challenged. Conventional approaches often are lacking when it comes to engaging students or providing practical application of programming concepts (Horlador, et., 2024). Traditionally, a beginner developer sits inside a classroom, listening to an instructor as they explain related programming concepts. However, there are instances wherein the beginner developer cannot follow along with the instructor's sample codes (Horlador, et., 2024).

The current education system is outdated in the sense that it cannot keep up with the demands of a fast-paced and information-rich world outside of the classroom. In response to these challenges, innovative learning platforms can be applied to transform the way beginner developers usually learn (Horlador, et., 2024).

In today's rapidly evolving landscape, mastering the concepts of Information Technology is crucial for the students aspiring to enter the technology industry (Abat, et al., 2024). Software development is fundamental to the innovation and efficiency of the applications and solutions tailored to specific needs and demands (Abat, et al., 2024). This caused the current curriculum to include programming languages such as Lua and Python (Abat, et al., 2024).

Despite the abundance of academic materials and tutorials regarding different types of programming languages, most of the students still face difficulties in learning the fundamentals of programming and how to apply them. By implementing interactive game-based learning, it allows students to learn using an alternative method of learning that is both fun and educational. (Abat et al., 2024).

Foreign Studies

According to Eltomi (2025), the use of Lua programming language allows the game developers to easily implement core gameplay mechanics such as character movements, physics, and collision detection. This study also identified the problems that game developers encountered such as no prior knowledge in Lua programming language, Game size and complexity, and bad game designs. In relation to the study, it explains the use of Lua programming language in game development. However, the proposed study lacks the idea of showing how the actual Lua language is used in game development.

In a study done by Verbovetsksvi (2025), The authors identified the basic requirements for a game application, such as relevance to educational objectives, incorporation of game mechanics, the ability to write and debug code in the application, individualized passing of the game, and the ability to report and rate learning outcomes of students. The paper contains some fragments of the game workspace and code of a gaming application.

Several studies compare the use of Python and Lua in game development, highlighting their distinct advantages. Python, with libraries like Pygame, is favored for its simplicity and extensive community support, making it ideal for rapid prototyping and educational game development (Oleksiuk et al., 2024). However, it tends to have slower performance compared to Lua. Lua is widely used in professional game engines due to its lightweight nature, fast execution, and seamless integration with C/C++ code, making it suitable for performance-critical game scripting (MoldStud, 2025).

Tyagi's study (2021) compares developing a 2D Flappy Bird game in both Python (Pygame) and Lua (LOVE engine), showing that while Python is easier for beginners, Lua provides better performance and embedding capabilities for commercial projects. Other research agrees that Lua excels in embedded scripting and resource-sensitive environments, whereas Python is often used for game prototyping, tooling, and educational purposes.

In conclusion, these studies demonstrate that the choice between Python and Lua depends on the project goals, with Lua being preferred for efficiency and embedding, and Python for ease of use and broad application.

Local Studies

According to a study done in the University of Mindanao-PALMA Cluster Campus Philippines aimed to determine the effectiveness of game-based learning, based on the findings it can be affirmed that implementing game-based learning strategies is highly effective, resulting in a notable enhancement of learners' academic performance compared to conventional teaching methods. Moreover, the results encourage teachers to revisit their instructional materials and integrate game-based learning materials aligned with specific learning objectives. Additionally, the study provides substantial evidence that game-based learning significantly enhances academic performance compared to traditional instructional resources (Cezar, 2024).

According to Cezar (2022), a study was done to measure the effect of digital game-based learning. Educators and authors argued that video games are constructed on well-developed theories to engage players. The gathered data were interpreted and analyzed; the results revealed that the experimental group achieved higher scores in their academic performance test. The findings showed a significant difference in the performance for students engaging in digital game-based learning.

According to Justo et al. (2022), a study was done to analyze students' perspectives of game-based learning. The results have been proven the game-based learning has a more significant influence on retention and class interaction when this technique is used. As a result, the findings say that game-based learning aids information acquisition and boosts student engagement through increased class interaction.

Foreign Systems

CodeCombat (2024) is a game-based learning platform that teaches languages such as Python through interactive, role-playing gameplay. Its custom-built coding engine and interpreter make it beginner friendly. The platform includes project-sharing features and teacher grading systems, making it usable in both personal and educational contexts. According to Ahlstrom (2025), the system not only simplifies coding, but they execute it in a method that emphasizes having fun. CodeCombat demonstrates how games can effectively serve as a medium for learning programming concepts.

OmegaCrafter (2025) is an open-world survival craft game set in an underdevelopment game world that is being attacked by a mysterious program. Grammi, the players' programmable sidekick can be programmed with various actions and tasks to help with mining, logging, moving materials, and crafting items (Preferred Networks, 2025)

The Farmer Was Replaced (2023) is a game for Python-like language to fully automate various farming tasks that would otherwise be time consuming. (Herzog, 2023). Unlike most programming games, it is not divided into distinct levels that players have to complete but instead features continuous progression. Programming is done in a simple language similar to Python. The game is designed to provide basic programming concepts to the player (Herzog, 2023).

Local Systems

Philippine Coding Camp (2023) is a digital learning center that teaches Filipino students how to code through structured, hands-on programs. The system provides learning courses which teach Lua as one of the foundations for beginner and younger students while also introducing Python later as a part of the game. In relation to the study, the system highlights accessibility in programming education while sustaining engagement and interactivity that traditional digital courses may not fully achieve.

Logiscool (2025) is a digital learning platform that focuses on coding and digital literacy education. Its courses integrate popular platforms to provide engaging environments to enhance coding concepts. The system demonstrates the effectiveness of combining fun and interactivity with education.

Hello World (2024) is a game that teaches the fundamentals of Java and C#. Players apply their skills into gameplay where challenges are introduced. The game progresses based on what the players have learned so far, from introduction to more advanced topics. It is a practical application of game-based learning in the growing industry of programming.

Synthesis

The reviewed foreign and local literature, studies, and systems collectively emphasize the growing importance of game-based learning in programming education. Foreign authors (Becker, 2022; Crudu, 2024; Viduka, 2021) highlighted how games foster immersion and focus, while also pointing to the industry relevance of programming languages such as Lua and Python. Local studies (Horlador et al., 2024; Abat et al., 2024) reinforced these ideas by showing how interactive, game-based methods can complement or even surpass traditional classroom instruction, especially in engaging students who struggle with conventional approaches. Moreover, both foreign and local systems (e.g., CodeCombat, 2024; Philippine Coding Camp, 2023) demonstrated practical applications of integrating coding into interactive platforms, providing evidence that game-based solutions are not only feasible but effective in real-world educational settings.

However, despite these advances, the reviewed literature also reveals a gap: while many systems teach Python or Lua through games, few explicitly focus on bridging the fundamentals of programming with practical coding immersion tailored for beginners in the Philippine context. This gap underscores the significance of the present study, which seeks to design a game-based solution that leverages the strengths of Lua and Python while addressing the challenges faced by novice developers in traditional learning environments. In the context of learning, both data gathered from foreign and local literature are that gaming tools have been used as a powerful supplementary tool for learning and can be applicable into learning complex topics such as concepts of programming (Becker, 2022; Horlador, et al., 2024).

The difference between the gathered information from foreign and local literatures are that in foreign literature, it has been found that games have been increasingly prominent in educational settings since when it comes to learning, students can be more focused and immersed (Becker, 2022). Local literature on the other hand has found that traditional methods are becoming outdated and are challenged by more innovative means of learning, especially considering the context of teaching complex topics in Computer Science (Horlador, et al., 2024).

Foreign studies done by authors (Eltomi, 2025; MoldStud, 2025; Tiyagi, 2021) gathered that Lua and Python are applicable in implementing features much easier due to each of the advantages that the two languages have. Lua is a lightweight language and integrates with C and C++ code while Python is more beginner-friendly and applicable for prototyping and tooling.

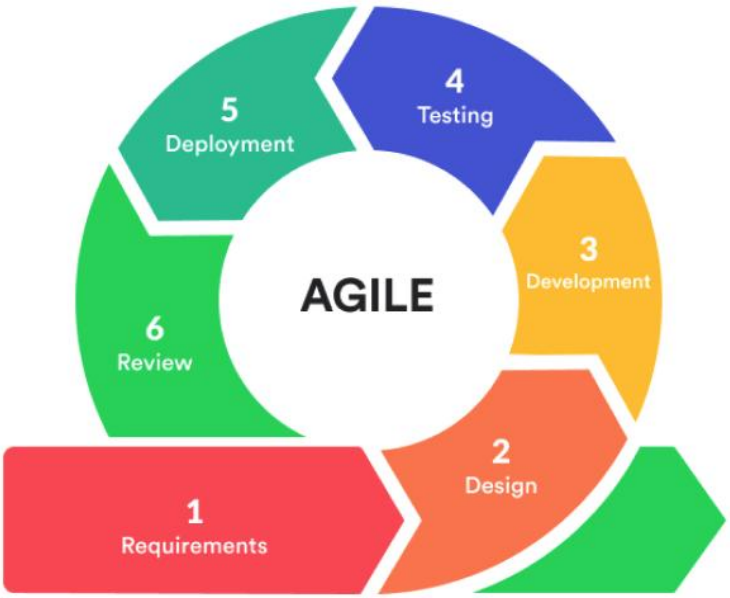
Local studies proven by authors (Cezar, 2022; Justo, et al., 2022) found that gamified learning is an effective method of learning as it has been gathered that the group that was engaged in gamified learning scored better in comparison to the group that did not. Based on what the study has gathered from the local and foreign studies, the goal is to implement gamified learning into teaching students the concepts of Lua and Python as these two languages have been shown to be appealing for beginners to learn (Oleksiuk, et al., 2024; MoldStud, 2025).

The proposed game aims to be unique by implementing the methods of gamified learning into teaching the concepts of Lua and Python in such a way that it acts as a supplement alongside traditional learning methods. It acts to teach students in a more engaging way. It does so by applying concepts and ideas that the foreign and local literature and studies have gathered and combine them in such a way that it introduces a specialized method of teaching students various programming concepts. Lua and Python were chosen as the two programming languages to teach because of their simplicity and utility, making them appealing to teach.

METHODOLOGY

Agile Methodology

The agile game development process consists of multiple steps that often occur simultaneously. It relies heavily on teamwork, with early and consistent communication among team members being essential to the project's success. Rather than following a long, sequential design process, agile web development adopts short, iterative cycles each with its own unique goal, timeline, and deliverables.



Goals

In this step, the team's goals are determined. During this phase, developers set the objective of creating a website capable of converting designs into code. The target users are identified, and the website's main functions undergo initial conceptual design.

Ideation

In this phase, the team collaborates to brainstorm innovative and creative ideas for the game. This stage focuses on generating possible features, exploring solutions, and identifying the most effective ways to implement them.

Planning

After gathering ideas, the team proceeds to create a structured plan. Tasks are prioritized, timelines are established, and responsibilities are assigned to each team member. The planning phase ensures that everyone understands the workflow, sprint duration, and deliverables for each cycle.

Sprint

The sprint phase is the core of the agile process. Here, the team works on specific project features or modules within a fixed timeframe—usually one to four weeks. At the end of each sprint, a functional component of the website is delivered, tested, and reviewed for improvement.

Iteration

Following each sprint, the team reviews feedback and refines the project. Adjustments, bug fixes, and feature enhancements are made based on testing results and stakeholder input. This iterative approach promotes continuous improvement and ensures that the product evolves toward meeting user needs effectively.

Production

Once all sprints and iterations are complete, the finalized game is deployed for production. The team conducts final testing, documentation, and optimization to ensure stability and performance. After deployment, maintenance and future updates are scheduled to address potential issues or feature requests.

Hardware/Software

Table 1. Hardware

Devices	Specifications
Desktop	CPU: AMD Ryzen 5 5600G Motherboard: MSI PRO B550M-P GEN3 GPU: Radeon Vega 7 Integrated Graphics RAM: 16 GB Operating System: CachyOS (Arch Linux)
Laptop	CPU: i5-12500h Motherboard:AN515-58-50YE GPU: RTX 3050M Ram: 8x2 DDR4 2400mhz sodimm Operating System: Windows 10 Pro

Table 2. Software

Software	Purpose
Godot Engine	Game Engine
GDScript	Game Logic
Krita	Art Editing Software
Rust	Performance critical implementation
Lua	Main Scripting Language
Python	Sub Scripting Language
VS Codium	Chosen IDE.
GitHub	Version Control Software, Logger, File Storage

Calendar of Activities

During the first week of September, the developers began working on the initial draft of the study's introduction and finalized the research title. This phase focuses on identifying the study's purpose, significance, and scope while aligning the objectives with the project's intended outcome. The week concluded with the team reviewing the overall direction of the research to ensure that it addressed relevant academic and industry needs.

In the second week of September, the team revised the introduction to improve clarity and alignment with the research objectives. They also began drafting Chapter I, which detailed the background of the problem, statement of the problem, and significance of the study. The group collaborated closely to ensure that the content presented a strong foundation for the succeeding chapters.

The third week of September was dedicated to additional revisions and refinements of Chapter I. The developers reviewed the document to ensure consistency, coherence, and adherence to academic standards. Feedback from peers and advisers was considered to enhance the quality and flow of the content, preparing the chapter for final review.

During the fourth week of September, the developers conducted the survey to gather data from the target respondents. The collected information served as the basis for understanding user preferences and needs. In the same week, the team began drafting Chapter II, which included the review of related literature, studies, and systems relevant to the proposed project.

The first week of October was devoted to extensive research on literature, existing studies, and software systems related to game-based learning. The developers consolidated and analyzed this information to strengthen Chapter II. By the end of the week, Chapter II was completed and prepared for initial revision.

In the second week of October, the proponents focused on drafting the hardware and software selection section. This included identifying the tools, technologies, and development environments necessary for system implementation. The week also involved outlining the technical specifications and requirements for the proposed application. During the third week of October, the team concentrated on revising and improving Chapter III, which presented the project's methodology. The developers refined the research design, data collection methods, and system development framework to ensure accuracy and alignment with research objectives. In the fourth week of October, the proponents carried out final revisions and grammar checks across all chapters to ensure consistency and correctness. The team also completed the initial design of the software user interface, marking the transition from the documentation phase to the beginning of system development.

Budgetary Estimate

Quantity	Specifics	Approximate Cost
1	Acer Nitro 5 AN515-50YE	PHP 49, 000
1 1 1 1 1	Ryzen 5 5600g Asus Prime B550 16 GB ram 1TB SSD 650w PSU	PHP 22, 295

Human Resources

Curriculum Vitae of

MAMAUAG, JEFFERSON G.

<Block 66 Lot 24 Nuvista Subdivision Barangay Santo Cristo City of San Jose Del Monte Bulacan>
mamauag.248577@sjdelmonte.sti.edu.ph
0931096819

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2023	STI College San Jose Del Monte
Vocational/Technical		
High School	May 2016-2023	Yverdon De Pestalozzi School
Elementary	June 2010-2016	Placido Del Mundo Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience Job Title	Name and Address of Company or Organization

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
July 2021-2023	Code Camp Society	Member
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	CS 2nd Year representative
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
MS Word	Intermediate	September 2016
Java Programming	Intermediate	December 2023
Python Programming	Intermediate	April 2025

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	National Service Training Program
February 2024	GitGud

Curriculum Vitae of

Tayag, Jose Mari Christian V.

Block 34, Lot 5, Saint Therese Phase, Deca Homes Subdivision, Barangay Loma de Gato, Marilao, Bulacan
tayag.334839@sjdelmonte.sti.edu.ph
+639166248648

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	**August 2023**	**STI College San Jose Del Monte**
Vocational/Technical		
High School	**May 2014-2020**	**Caloocan National Science and Technology Highschool**
Elementary	**June 2008-2014**	**ST. Anne Power Academy-Marilao, inc.**

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	Member
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Ux/Ui Design	Intermediate	September 2020
Python Programming	Intermediate	March 2018
Rust Programming	Intermediate	December 2024

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	**National Service Training Program**
February 2024	**GitGud**
	CodeFest

Curriculum Vitae of

Accad, Glenn Marco B.

Block 32, Lot 17 Phase 6C Towerville, Barangay Gaya - Gaya, Towerville, San Jose Del Monte Bulacan
Accad.355761@sjdelmonte.sti.edu.ph
+639981847987

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2023	STI College San Jose Del Monte
Vocational/Technical		
High School	May 2017-2023	Graceville National High school
Elementary	June 2011-2017	Marangal Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience Job Title	Name and Address of Company or Organization
February 2023 – May 2023	Student Assistant	San Jose Academy of Bulacan

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	Member
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
MS Office	Intermediate	June 2011
Java Programming	Intermediate	October 2023
C# Programming	Intermediate	May 2024

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	**National Service Training Program**
February 2024	**GitGud**
	CodeFest

Curriculum Vitae of

Gallano, Adrian V.

1067 Malaya Street, Sapang Alat Road, Barangay 181, Pangarap Village, Caloocan City, NCR
gallano.360632@sjdelmonte.sti.edu.ph
+63927208694

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2023	STI College San Jose Del Monte
Vocational/Technical		
High School	July 2017-2023	Immaculada Conception Colleges
Elementary	August 2011-2017	Our Lady of Mercy School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience Job Title	Name and Address of Company or Organization

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	Member
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
MS Office	Intermediate	June 2018
Java Programming	Intermediate	October 2023
C# Programming	Intermediate	September 2024

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

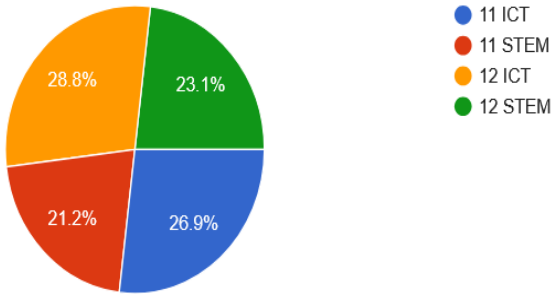
Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	National Service Training Program
February 2024	GitGud

Survey

What is your academic strand?

Copy chart

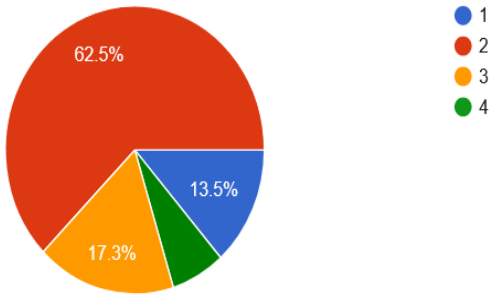
104 responses



In a scale of 1-4, how familiar are you with the Lua programming language?

Copy chart

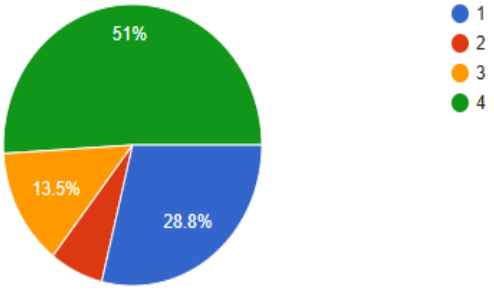
104 responses



In a scale of 1–4, how easy do you find Lua's structure and syntax?

Copy chart

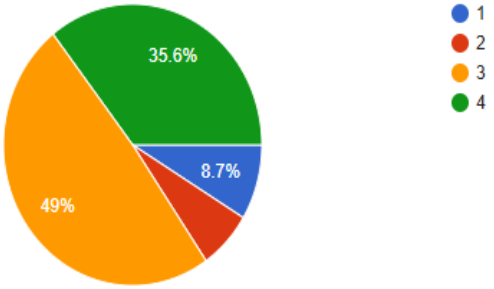
104 responses



In a scale of 1–4, how confident would you feel using Lua for real-world problems?

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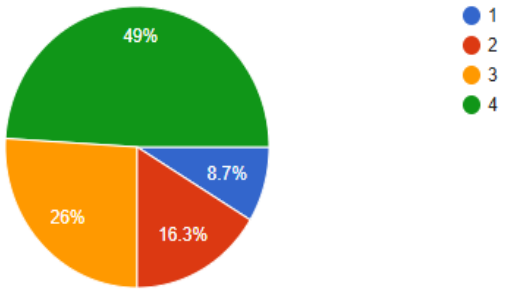
104 responses



In a scale of 1–4, to what extent would learning Lua through game programming help you?

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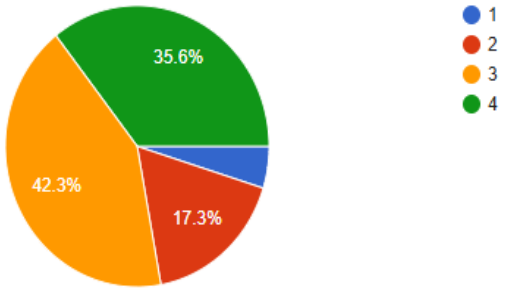
104 responses



In a scale of 1–4, how appealing do you find Lua as a new skill to learn?

Copy chart

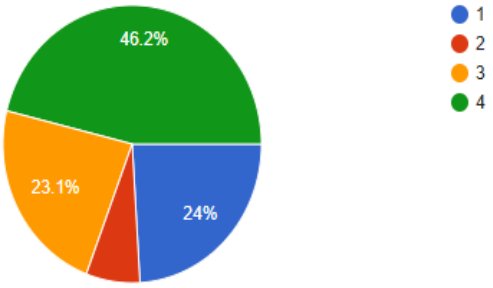
104 responses



In a scale of 1–4, how familiar are you with survival games?

Copy chart

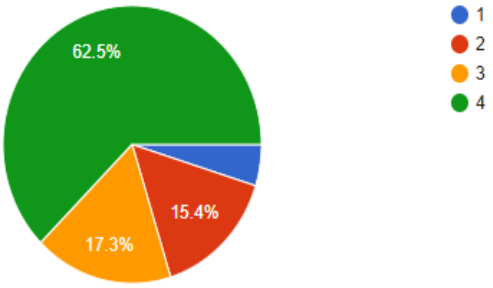
104 responses



In a scale of 1–4, how motivated would you be to learn Lua using a game?

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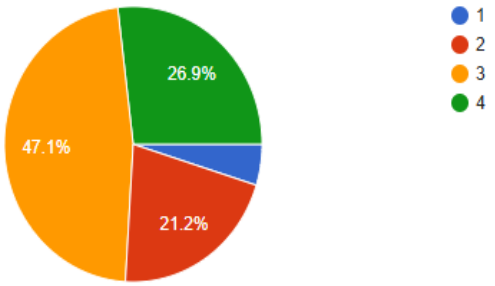
104 responses



In a scale of 1–4, how likely are you to experiment using Lua inside a game?

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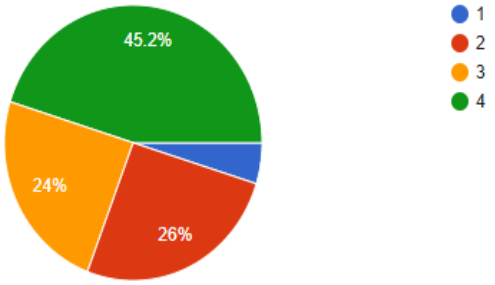
104 responses



In a scale of 1–4, how familiar are you with the Python programming language?

Copy chart

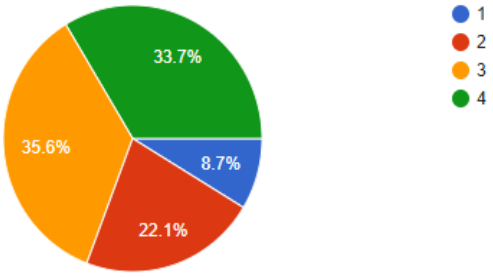
104 responses



In a scale of 1–4, how easy do you find Python's structure and syntax?

Copy chart

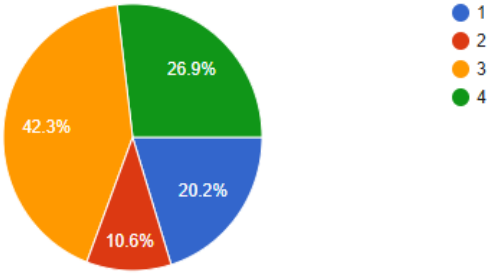
104 responses



In a scale of 1–4, to what extent would learning Python through game programming help you?

Copy chart

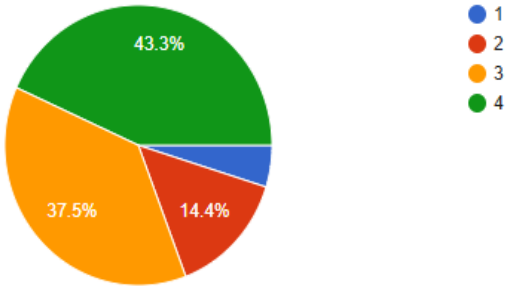
104 responses



In a scale of 1–4, how appealing do you find Python as a new skill to learn?

Copy chart

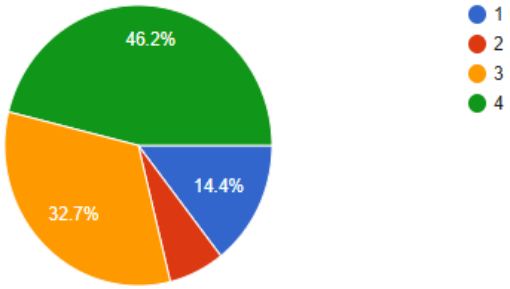
104 responses

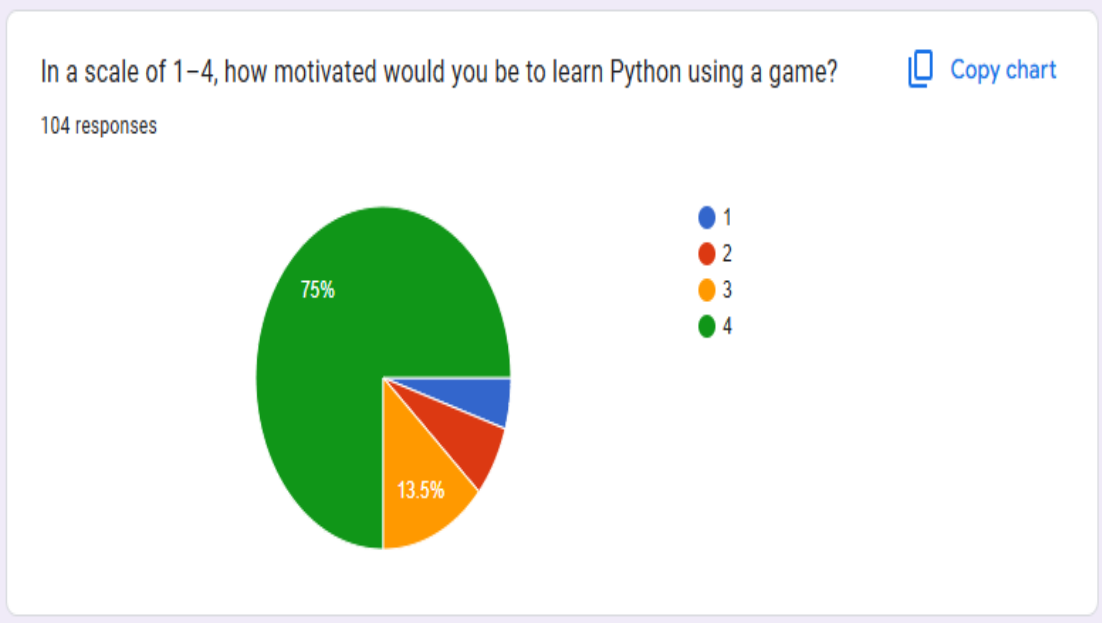


In a scale of 1–4, how confident would you feel learning Python for basic programming tasks?

Copy chart

104 responses




In a scale of 1–4, how motivated would you be to learn Python using a game?
Copy chart
104 responses
75%
13.5%
1
2
3
4

ADVISER'S ACCEPTANCE FORM

NAME OF PROPONENTS:	**Glenn Marco B. Accad** **Adrian V. Gallano** **Jefferson G. Mamauag** **Jose Mari Christian V. Tayag**
APPROVED RESEARCH TITLE:	**SandEE: An Educational 2D Open World Survival Crafting Game for Lua and Python**
AREA OF STUDY:	**Cross-Platform Application, Game-Based Learning, Programming**

CONFORME:

Norvin Shaq F. Lintag
Thesis Adviser

APPROVED BY:

Enoch James C. Tayanguna **Date: 13 September 2025**
Thesis Coordinator

NOTED BY:

Deffrey Josh A. Austria
Program Head

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